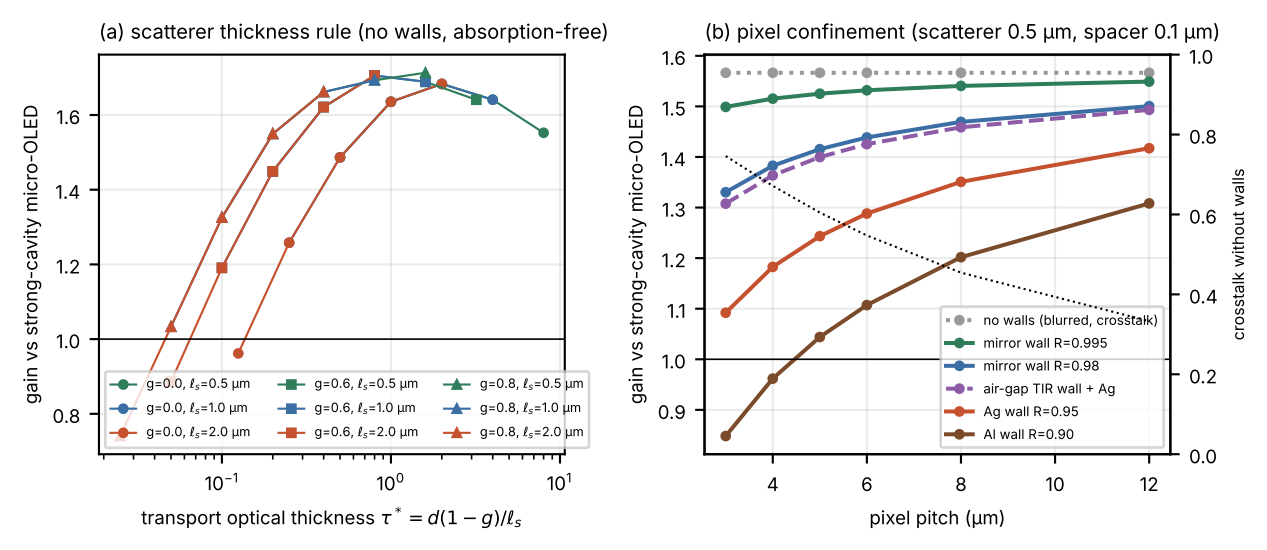
(a) scatterer thickness rule (no walls, absorption-free)
gain vs strong-cavity micro-OLED
transport optical thickness $\tau^* = d(1-g)/\ell_s$
g=0.0, ℓ_s=0.5 µm
g=0.0, ℓ_s=1.0 µm
g=0.0, ℓ_s=2.0 µm
g=0.6, ℓ_s=0.5 µm
g=0.6, ℓ_s=1.0 µm
g=0.6, ℓ_s=2.0 µm
g=0.8, ℓ_s=0.5 µm
g=0.8, ℓ_s=1.0 µm
g=0.8, ℓ_s=2.0 µm
(b) pixel confinement (scatterer 0.5 µm, spacer 0.1 µm)
gain vs strong-cavity micro-OLED
crosstalk without walls
pixel pitch (µm)
no walls (blurred, crosstalk)
mirror wall R=0.995
mirror wall R=0.98
air-gap TIR wall + Ag
Ag wall R=0.95
Al wall R=0.90